

Development of SMD Solvent Descriptors for Supercritical Methanol and Ethanol: Thermodynamic Analysis of Biomass-Derived Reactions

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01 INTRODUCTION

Lignin-derived compounds are promising renewable feedstocks for the production of value-added chemicals. Supercritical alcohols have emerged as attractive reaction media; however, the influence of their solvent properties on the thermodynamic feasibility of lignin-derived reactions remains poorly understood. In this work, solvent descriptors of supercritical methanol and ethanol were determined and used to evaluate the Gibbs free energy (ΔG) and enthalpy (ΔH) of two representative lignin-derived conversion reactions. The calculated thermodynamic parameters were compared with those in supercritical water to assess the effect of solvent on reaction feasibility.

02 OBJECTIVES

- ✓ Determine the solvent descriptors of supercritical methanol and ethanol.
- ✓ Conducting the thermodynamic analysis of two lignin-derived reactions.
- ✓ Compare the thermodynamic parameters with supercritical water.
- ✓ Evaluate the effect of solvent on reaction thermodynamics and feasibility.

03 METHODOLOGY

Solvent descriptors for supercritical methanol and ethanol were generated using density from the CoolProp library, refractive indices calculated by the Lorentz–Lorenz equation, and experimental Abraham hydrogen-bond acidity (α) and basicity (β) parameters. DFT calculations were performed in Gaussian using the SMD solvation model with the 6-311++G(d,p) basis set, including geometry optimization, frequency, and intrinsic reaction coordinate (IRC) analyses. Thermodynamic parameters, including Gibbs free energy (ΔG) and enthalpy (ΔH), were evaluated at four supercritical temperature–pressure conditions (Methanol: 533 K/98.69 atm, 543 K/204.3 atm, 553 K/204.3 atm, and 593 K/205.2 atm; Ethanol: 553 K/107.6 atm, 573 K/108.6 atm, 593 K/224.1 atm, and 613 K/231.9 atm) and compared with supercritical water.

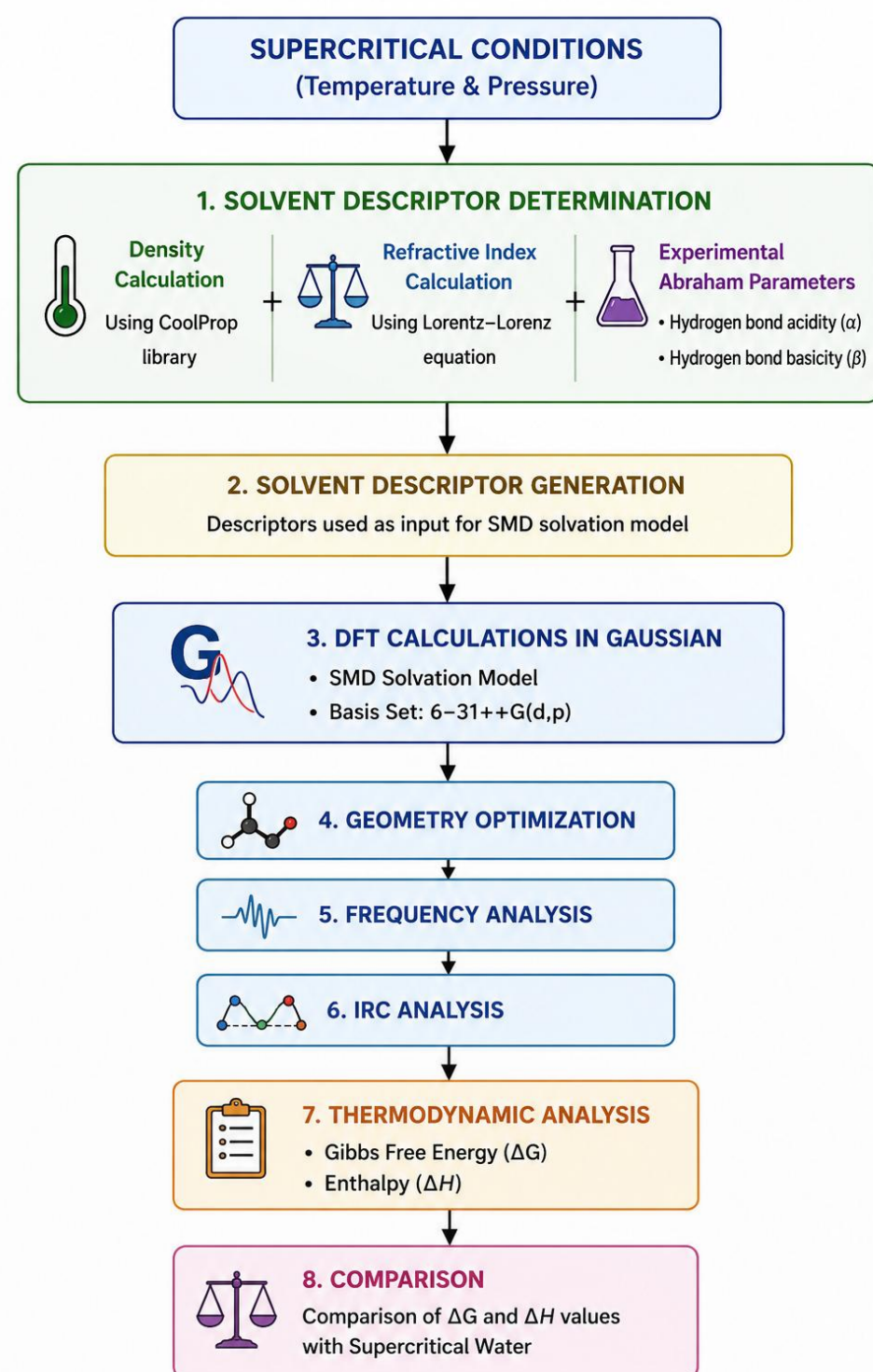


Figure 1. Schematic of the proposed methodology.

04 RESULTS

Negative ΔG values confirmed the thermodynamic feasibility of both lignin-derived reactions under all supercritical conditions. The guaiacol $\rightarrow$ cyclohexanone pathway exhibited a stronger thermodynamic driving force than the ferulic acid $\rightarrow$ cinnamic acid pathway. Negative ΔH values indicated exothermic behavior in both supercritical methanol and ethanol.

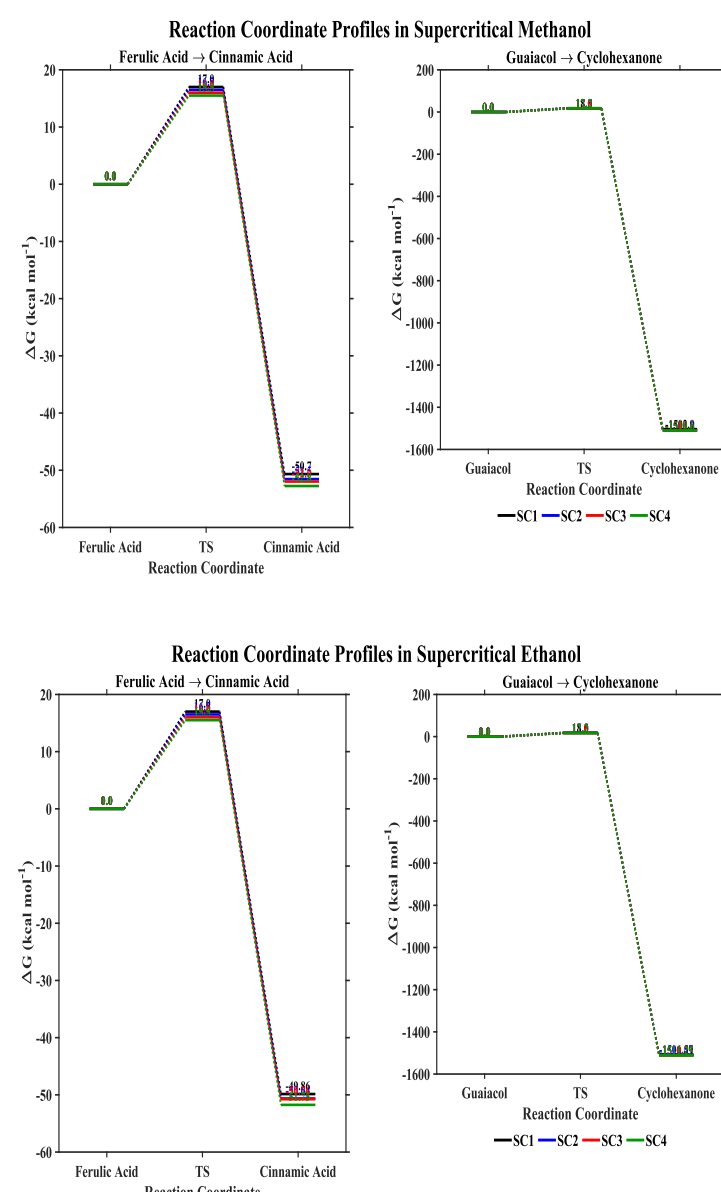


Figure 2. Reaction coordinate diagrams illustrating the Gibbs free energy (ΔG) profiles for the conversion of ferulic acid to cinnamic acid and guaiacol to cyclohexanone under four supercritical operating conditions (SC1–SC4) in supercritical methanol and supercritical ethanol.

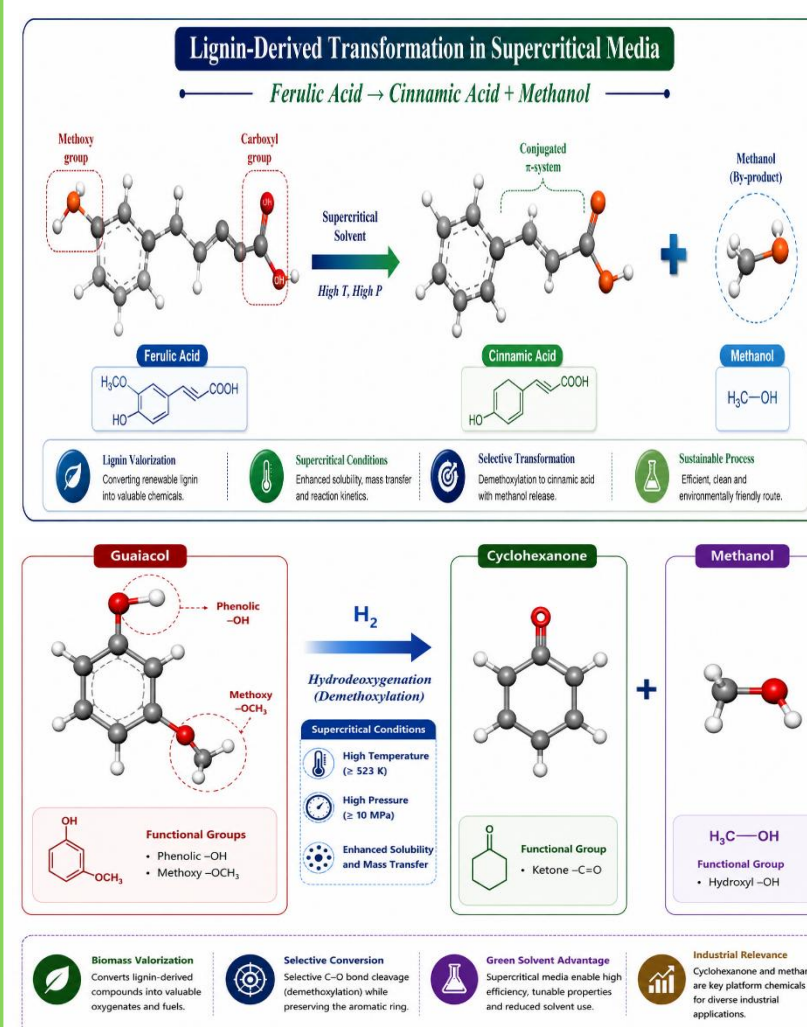


Figure 3. Selected lignin-derived reaction mechanisms investigated in this study: conversion of ferulic acid to cinnamic acid and conversion of guaiacol to phenol and methanol.

05 DISCUSSION

Negative ΔG values confirmed the thermodynamic favorability of both lignin-derived conversion pathways. Supercritical methanol and ethanol generally produced more negative ΔG values than supercritical water, indicating a stronger thermodynamic driving force in alcohol-based media. The negative ΔH values observed for the investigated reactions indicate exothermic behavior. Overall, the results demonstrate the significant influence of the supercritical solvent environment on lignin conversion thermodynamics.

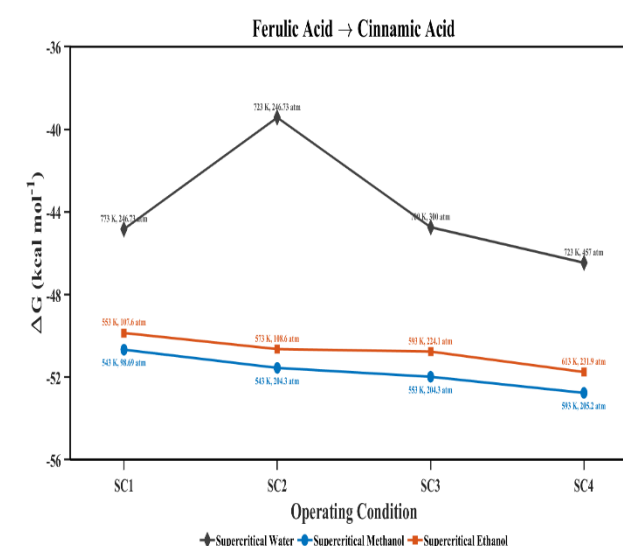


Figure 4. Comparison of Gibbs free energy (ΔG) for ferulic acid $\rightarrow$ cinnamic acid in supercritical water, methanol, and ethanol.

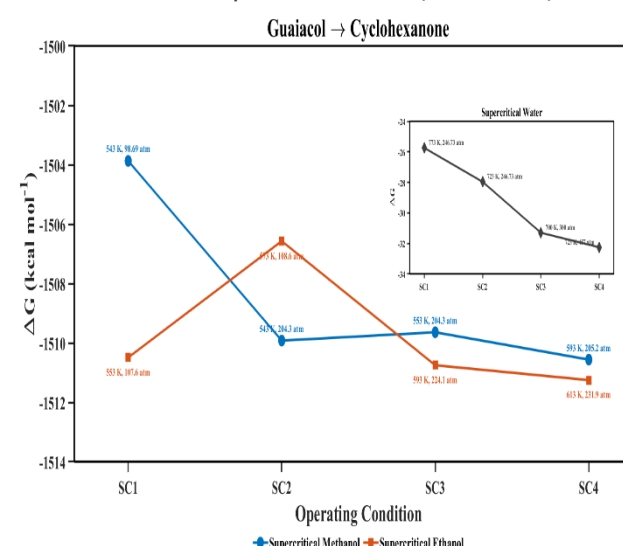


Figure 5. Comparison of Gibbs free energy (ΔG) for guaiacol $\rightarrow$ cyclohexanone in supercritical water, methanol, and ethanol.

06 CONCLUSIONS

- Both reactions were thermodynamically favorable under supercritical alcohol conditions.
- Supercritical methanol and ethanol influenced the reaction energetics differently, demonstrating the importance of solvent selection.
- The findings provide insights for designing sustainable lignin valorization processes and renewable chemical production.

07 FUTURE WORK

Future work will focus on reaction kinetics, experimental validation, and scale-up of lignin valorization under supercritical conditions. Further lignin-derived reaction pathways and solvent systems will also be explored.

08 REFERENCES

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